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# Field Case Study Report: «Avare Biotech» App Powered by MAKSA technology Validation on the Frozen–Thawed semen (Pune, INDIA, 2025)

Date: December 2025
Location: Pune, INDIA

| | |
|---|---|
| Conducted by:<br><br>Dr. Kaustubh Vishram Natu (Pune, INDIA) | Verified by:<br><br>Avare Biotech Team (Dubai, UAE) |
| **Head Veterinarian at NATSON VETERINARY SERVICES**<br><br>**(PUNE, INDIA)** | **Maksim Mershiev**<br><br>The CEO&Co-Founder of Avare Biotech |
| | |
| Signature: Signed by: 0DB009887F7E43D... | Signature: Signed by: Maksim Mershiev 325996BF0B2B43A... |
| 12/26/2025 | 12/26/2025 |

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## Executive Summary / Conclusions

- The smartphone-based MAKSA technology proved to be simple and intuitive to use for both fresh and frozen-thawed bovine semen samples, allowing rapid on-site evaluation.
- Accuracy differences for concentration and progressive motility remained below the 10% threshold, indicating suitability for field use.
- Overall, the experience of working with the Avare Biotech app powered by MAKSA technology was excellent, demonstrating strong potential for routine field use and further technical refinement in India.

## Objective

- To validate the smartphone-based MAKSA (Mobile-Assisted Key Semen Analysis) technology powering Avare Biotech App for **frozen bull semen** evaluation and comparison with manual counting methods.

## Materials and Methods

- **According to the given protocol**, a comparison was made between MAKSA and Manual techniques to evaluate Concentration, Total and Progressive motility percentages.
- **All the basic routine procedures including thawing, slide preparation, microscope adjustment etc., were done systematically.**
- As per the suggestion in the webinar, 3-4 sec. video of frozen-thawed semen was taken from 5 different fields of the same sample.
- Visual assessment was done for the manual count of Total and Progressive motility percentages.

- Afterwards **Manual count of sperm** was done as per the guidelines given by the following procedure and formula.

https://www.ansci.wisc.edu/jjp1/ansci_repro/lab/procedures/hemacytometer/Hemocytometer%20use.html

$$\textbf{Concentration/ml} = \textbf{(Dilution Factor)(Count in 5 squares)}(0.05 \times 10^{6})$$

For better visibility, 10 microliters of the semen sample was diluted in 90 microliters of Normal Saline to obtain 1:10 dilution. (Dilution factor =10)

*Table 1 - MAKSA vs MANUAL tests results comparison*

| Sample (Bull/Straw) | Parameters/Method | Avare App powered by MAKSA tech | Manual Count | Difference, % | Notes |
|---|---|---|---|---|---|

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| | | | | | |
|---|---|---|---|---|---|
| HFCB 50182 | Concentration (M/mL) | 47.36 | 45 | **5.11** | |
| | Total Motility (%) | | 81 | | |
| | Progressive Motility (%) | 82 | 76.6 | **6.81** | |
| | | | | | |
| JY 50065 | Concentration (M/mL) | 51.52 | 58 | **11.84** | |
| | Total Motility (%) | | 75 | | |
| | Progressive Motility (%) | 76.2 | 71.8 | **5.95** | |
| | | | | | |
| MR 50424 | Concentration (M/mL) | 59.52 | 59 | **0.88** | |
| | Total Motility (%) | | 39 | | |
| | Progressive Motility (%) | 65.8 | 28 | **80.6** | |
| | | | | | |

## Discussion / Expert Insights

- A validation trial was conducted to assess the performance of the MAKSA in comparison with the standard manual method for semen analysis. The results demonstrated a **good correlation between sperm concentration per milliliter obtained manually and those generated by the MAKSA**, indicating that the application reliably estimates sperm concentration. Additionally, MAKSA produced **consistent results with minimal variation across repeated measurements**, supporting its repeatability and reliability.

- IN case of MURRAH BUFFALO, During the evaluation of motility parameters, it was observed that **movement of the background fluid on the semen slide influenced motility estimation**, leading to errors in motility counts generated by MAKSA. This issue particularly

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affected the accurate differentiation of motile categories and requires technical refinement to minimize the impact of background motion.

- Further, although **progressive and non-progressive motility percentages should ideally show a logical correlation**, some inconsistency was noted, suggesting the need for improved classification algorithms. Despite these limitations, the software showed **reliable performance in assessing sperm concentration and progressive motility**, making it a trustworthy tool for these parameters.

## Recommendations

- Overall, MAKSA demonstrates strong potential for routine use in semen analysis, particularly for **concentration and progressive motility assessment**.
- Future improvements should focus on **correcting background motion artifacts**, **refining motility classification**, and **strengthening the correlation between progressive and non-progressive motility percentages** to enhance overall accuracy.

## Appendix

• MAKSA PDF reports

• Manual count results

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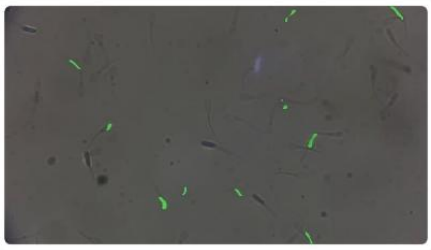
Bull Information

- Breed – Holstein Friesian

| | |
|---|---|
| INAPH ID | RHR-HX-50182 |
| BULL NO | HX-50182 |
| Date of Birth (DD-MM-YY) | 21-08-16 |
| Category | CATTLE |
| Breed | HFCB |
| Dam No. / Name | 95266147 |
| Dam's Best Lact.Yield (Kg) | 6198 |
| Fat % | 4.1 |
| Sire No./ Name | HF-158 |
| Sire's Dam's Best Lact. Yield (Kg) | 9280 |
| Sire Daughters Yield | NA |
| Breeding Value | NA |
| Star Value | 1* |

**Analysis 17.12.2025**

**Animal:** — **Name:** HFC1 HFCB50182 **Age:** 8 **Breed:** HF **Status:** Ready **Created:** 2025-12-17 13:03



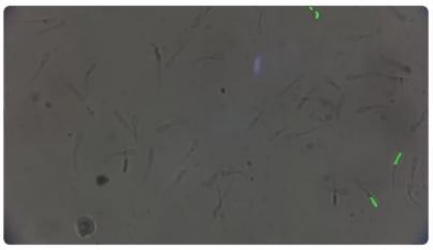
| | |
|---|---|
| Total objects: | 22 |
| Concentration (millions/ml^3): | 35.20 |
| Dilution rate: | [1:0] |
| Volume of field of view (mm^3): | 0.002179 |
| Progressive Motility: | 82% |
| Non-Progressive Motility: | 60% |
| Immotility: | 1% |

**Agglutination**

| | |
|---|---|
| Found: | No |

**Analysis 17.12.2025**

**Animal:** — **Name:** HFC2 HFCB 50182 **Age:** 8 **Breed:** HF **Status:** Ready **Created:** 2025-12-17 13:06



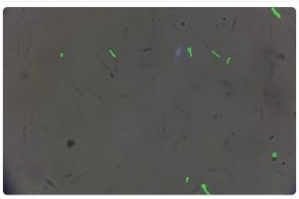
| | |
|---|---|
| Total objects: | 28 |
| Concentration (millions/ml^3): | 44.80 |
| Dilution rate: | [1:0] |
| Volume of field of view (mm^3): | 0.001786 |
| Progressive Motility: | 84% |
| Non-Progressive Motility: | 59% |
| Immotility: | 1% |

**Agglutination**

| | |
|---|---|
| Found: | No |

**Analysis 17.12.2025**

**Animal:** — **Name:** HFC3 HFCB50182 **Age:** 8 **Breed:** HF **Status:** Ready **Created:** 2025-12-17 13:08



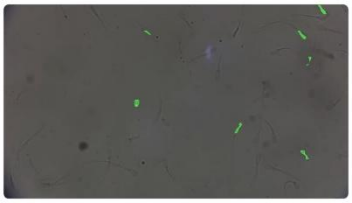
| | |
|---|---|
| Total objects: | 12 |
| Concentration (millions/ml^3): | 19.20 |
| Dilution rate: | [1:0] |
| Volume of field of view (mm^3): | 0.001203 |
| Progressive Motility: | 76% |
| Non-Progressive Motility: | 50% |
| Immotility: | 1% |

**Agglutination**

| | |
|---|---|
| Found: | No |

**Analysis 17.12.2025**

**Animal:** — **Name:** HFC4 HFCB50182 **Age:** 8 **Breed:** HF **Status:** Ready **Created:** 2025-12-17 15:09



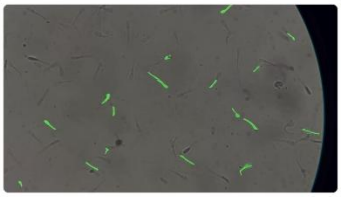
| | |
|---|---|
| Total objects: | 32 |
| Concentration (millions/ml^3): | 51.20 |
| Dilution rate: | [1:0] |
| Volume of field of view (mm^3): | 0.001420 |
| Progressive Motility: | 86% |
| Non-Progressive Motility: | 52% |
| Immotility: | 1% |

**Agglutination**

| | |
|---|---|
| Found: | No |

**Analysis 17.12.2025**

**Animal:** — **Name:** HFC5 HFCB 50182 **Age:** 8 **Breed:** HF **Status:** Ready **Created:** 2025-12-17 15:26



| | |
|---|---|
| Total objects: | 54 |
| Concentration (millions/ml^3): | 86.40 |
| Dilution rate: | [1:0] |
| Volume of field of view (mm^3): | 0.001934 |
| Progressive Motility: | 82% |
| Non-Progressive Motility: | 53% |
| Immotility: | 1% |

**Agglutination**

| | |
|---|---|
| Found: | No |

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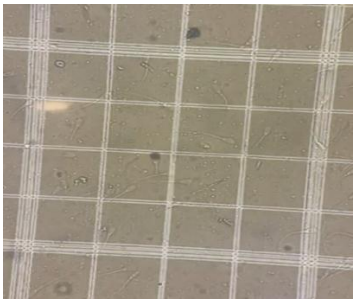
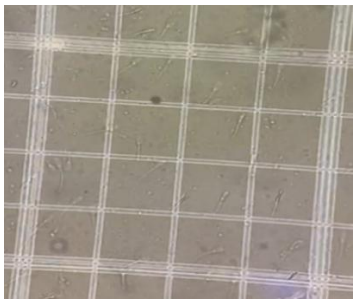
**Concentration (M/mL)**

Concentration (M/mL) by Manual method

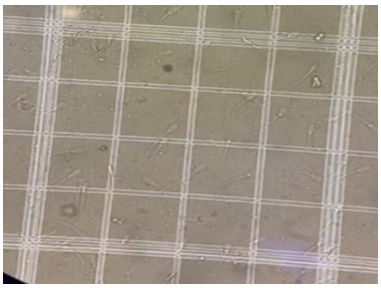


HF1

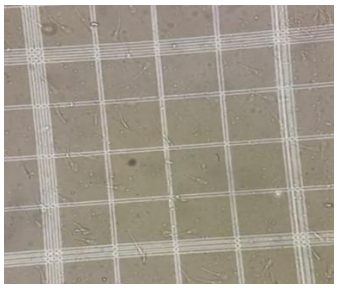
HF2



HF3



HF4



HF5

As per the formula-
Concentration per ml = (Dilution factor) × (count in 5 squares) × ($0.05 \times 10^6$)

=(10) X (16+21+19+19+15) X ($0.05 \times 10^6$)
=10 X 90X 50000
=**45 (M/mL)**

Comparison between Avare App powered by MAKSA tech and Manual method
**For Concentration (M/mL)**

| Sample No | Concentration (M/mL) by MAKSA | Concentration (M/mL) by the Manual method |
|---|---|---|
| HFC1 | 35.2 | |

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| | | |
|---|---|---|
| HFC2 | 44.8 | |
| HFC3 | 19.2 | |
| HFC4 | 51.2 | |
| HFC5 | 86.4 | |
| **Average** | **47.36** | **45** |

Comparison between Avare App powered by MAKSA tech and Manual method for **Progressive Motility (%)**

| Sample No | Progressive Motility (%) by MAKSA | Progressive Motility (%) by Manual method |
|---|---|---|
| HFC1 | 82 | 74 |
| HFC2 | 84 | 76 |
| HFC3 | 76 | 75 |
| HFC4 | 86 | 80 |
| HFC5 | 82 | 78 |
| **Average** | **82** | **76.6** |

| Sample No | **Total Motility (%)** by Manual method |
|---|---|
| HFC1 | 80 |
| HFC2 | 80 |
| HFC3 | 80 |
| HFC4 | 85 |
| HFC5 | 80 |
| **Average** | **81** |

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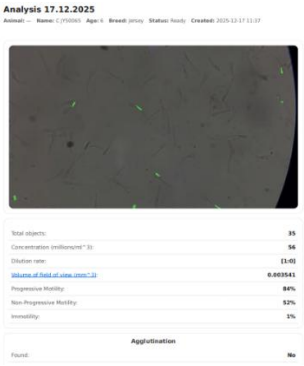
## Bull Information

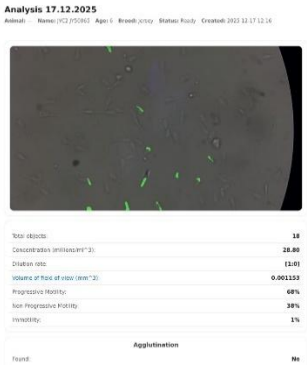
- Breed – Jersey

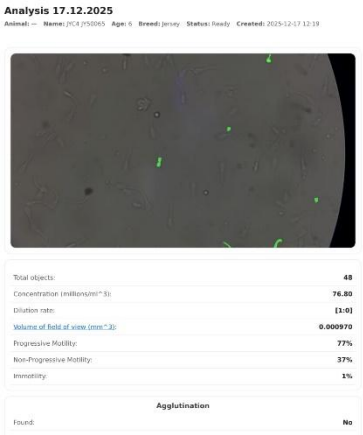
-

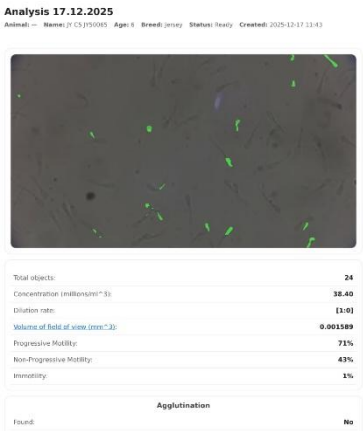
-

| | |
|---|---|
| INAPH ID | RHR-JY-50065 |
| BULL NO | JY-50065 (HELLFIRE) |
| Date of Birth (DD-MM-YY) | 27-03-15 |
| Category | CATTLE |
| Breed | Jersey |
| Dam No. / Name | 019903-04817 |
| Dam's Best Lact.Yield (Kg) | 7014 |
| Fat % | 6.74 |
| Sire No./ Name | VJ LINK |
| Sire's Dam's Best Lact. Yield (Kg) | 8311 |
| Sire Daughters Yield | N/A |
| Breeding Value | NA |
| Star Value | 1* |





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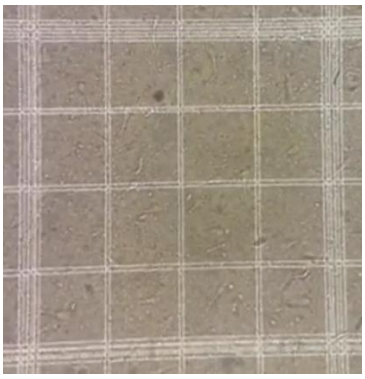
**Concentration (M/mL)**

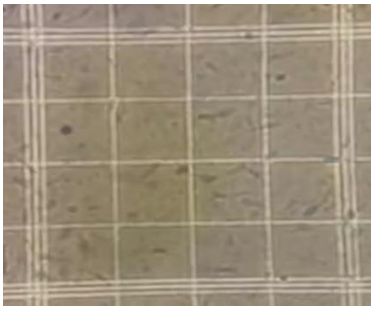
Concentration (M/mL) by Manual method



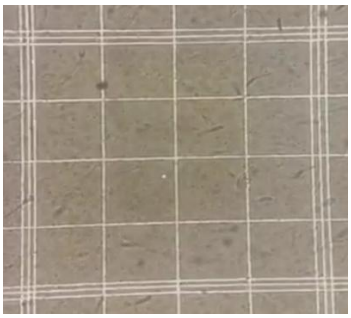
J1

J2

J3



J4



J5

As per the formula-
Concentration per ml = (Dilution factor) × (count in 5 squares) × ($0.05 \times 10^6$)

=(10) X (22+24+24+24+22) X ($0.05 \times 10^6$)
=10 X 116 X 50000
=**58 (M/mL)**

Comparison between Avare App powered by MAKSA tech and Manual method
**For Concentration (M/mL)**

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| Sample No | Concentration (M/mL) by MAKSA | Concentration (M/mL) by the Manual method |
|---|---|---|
| JYC1 | 56 | |
| JYC2 | 28.8 | |
| JYC3 | 57.6 | |
| JYC4 | 76.8 | |
| JYC5 | 38.4 | |
| **Average** | **51.52** | **58** |

Comparison between Avare App powered by MAKSA tech and Manual method For **Progressive Motility (%)**

| Sample No | Progressive Motility (%) by MAKSA | Progressive Motility (%) by Manual method |
|---|---|---|
| JYC1 | 84 | 78 |
| JYC2 | 68 | 65 |
| JYC3 | 81 | 78 |
| JYC4 | 77 | 72 |
| JYC5 | 71 | 66 |
| **Average** | **76.2** | **71.8** |

| Sample No | Total Motility (%) by Manual method |
|---|---|
| JYC1 | 80 |
| JYC2 | 70 |
| JYC3 | 80 |
| JYC4 | 75 |
| JYC5 | 70 |
| **Average** | **75** |

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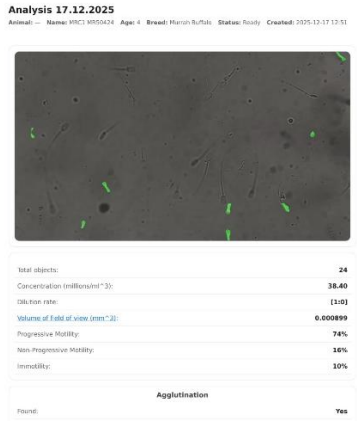
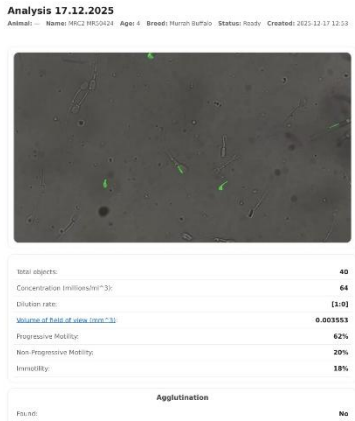
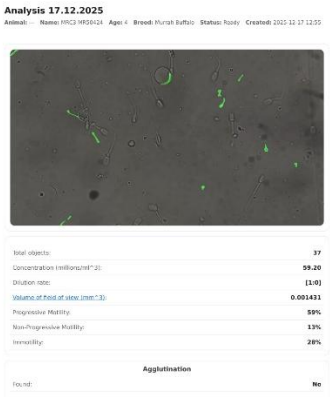
## Bull Information

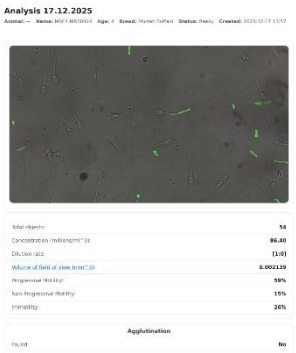
- Breed – Murrah Bull

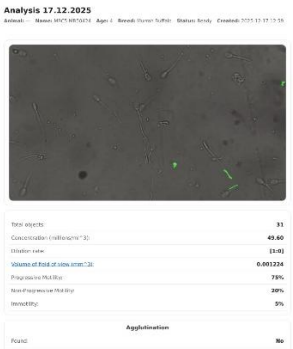
| | |
|---|---|
| INAPH ID | RAH-MR-50424 |
| BULL NO | MR-50424 |
| Date of Birth (DD-MM-YY) | 04-08-21 |
| Category | BUFFALO |
| Breed | Murrah |
| Dam No. / Name | 130006352284 |
| Dam's Best Lact.Yield (Kg) | 3617 |
| Fat % | 7.5 |
| Sire No./ Name | ROH-RM-013 |
| Sire's Dam's Best Lact. Yield (Kg) | NA |
| Sire Daughters Yield | NA |
| Breeding Value | 48 |
| Star Value | 1* |



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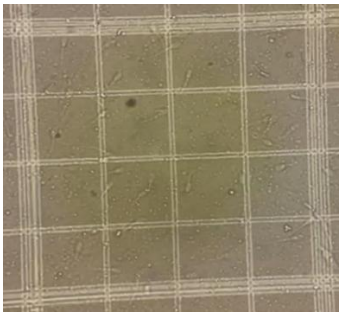
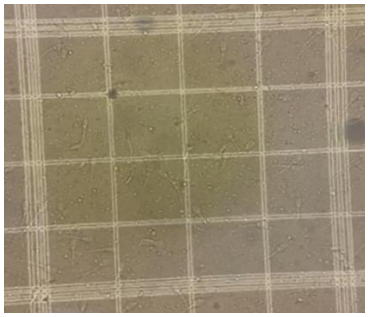
**Concentration (M/mL)**

Concentration (M/mL) by Manual method



M1



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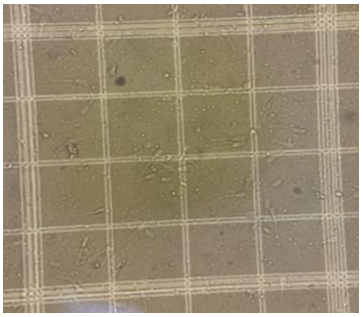

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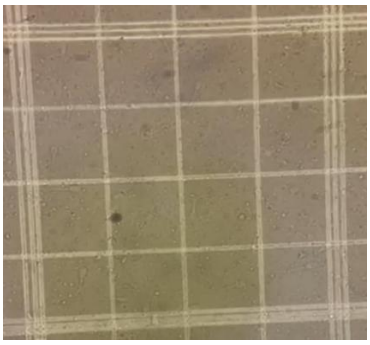
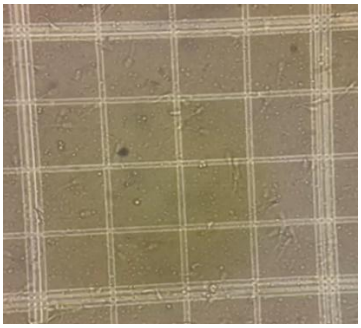


M4



M5

As per the formula-

Concentration per ml = (Dilution factor) × (count in 5 squares) × ($0.05 \times 10^6$)

=(10) X (28+30+24+13+23) X ($0.05 \times 10^6$)

=10 X 118 X 50000

=**59 (M/mL)**

Comparison between Avare App powered by MAKSA tech and Manual method

**For Concentration (M/mL)**

Docusign Envelope ID: D4F3A451-5809-420D-9D5B-9F5546D7757F



| Sample No | Concentration (M/mL) by MAKSA | Concentration (M/mL) by the Manual method |
|---|---|---|
| MRC1 | 38.4 | |
| MRC2 | 64 | |
| MRC3 | 59.2 | |
| MRC4 | 86.4 | |
| MRC5 | 49.6 | |
| **Average** | **59.52** | **59** |

Comparison between Avare App powered by MAKSA tech and Manual method For **Progressive Motility (%)**

| Sample No | Progressive Motility (%) by MAKSA | Progressive Motility (%) by Manual method |
|---|---|---|
| MRC1 | 74 | 30 |
| MRC2 | 62 | 10 |
| MRC3 | 59 | 20 |
| MRC4 | 59 | 30 |
| MRC5 | 75 | 50 |
| **Average** | **65.8** | **28** |

| Sample No | Total Motility (%) by Manual method |
|---|---|
| MRC1 | 35 |
| MRC2 | 10 |
| MRC3 | 40 |
| MRC4 | 50 |
| MRC5 | 60 |
| **Average** | **39** |